

MULTI-SOURCE PLAGIARISM TEST DOCUMENT

(WITH TEXT AND IMAGES)

This document combines content from TWO different source papers:

Paper A: Chromatin-lamin B1 interaction promotes genomic compartmentalization

DOI: uuid_0bde5539-6c0d-1014-8c2e-b50b932a95b0

Paper B: H3.3 is essential for gastrulation of *Xenopus laevis*

DOI: uuid_0ba6a324-6c1f-1014-b700-b93a8e0baef4

Expected Result: Plagiarism detection should identify TEXT and IMAGE matches from BOTH sources.

SECTION 1: CONTENT FROM PAPER A

Source: uuid_0bde5539-6c0d-1014-8c2e-b50b932a95b0

Chromatin-lamin B1 interaction promotes genomic compartmentalization and constrains chromatin dynamics

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Abstract

The eukaryotic genome is folded into higher-order conformation accompanied with constrained dynamics for coordinated genome functions. However, the molecular machinery underlying these hierarchically organized chromatin architecture and dynamics remains poorly understood. Here by combining imaging and Hi-C sequencing, we studied the role of lamin B1 in chromatin architecture and dynamics. We found that lamin B1 depletion leads to chromatin redistribution and decompaction. Consequently, the inter-chromosomal interactions and overlap between chromosome territories are increased.

Moreover, Hi-C data revealed that lamin B1 is required for the integrity and segregation of chromatin compartments but not for the topologically associating domains (TADs). We further proved that depletion of lamin B1 leads to increased chromatin dynamics, owing to chromatin decompaction and redistribution toward nuclear interior.

Introduction

Chromatin in the interphase nucleus of eukaryotic cells are highly compartmentalized and structured. Owing to technological breakthroughs in imaging and sequencing, chromatin higher-order structure has been increasingly studied over the last decade. Hierarchical chromatin architecture is composed of loops, TADs, active and inactive A/B compartments, and chromosome territories, in increasing scales.

Nuclear lamina consists of many protein complexes, and lamins are the main components of nuclear lamina in most mammalian cells and can be classified into A- and B-type lamins. Lamin B1 mainly localizes at the nuclear periphery, while A-type lamins are also found in the nucleoplasm.

Results

Lamin B1 depletion leads to chromatin redistribution and decompaction. To explore the potential role of lamin B1 in nuclear chromatin organization, we first investigated the subnuclear distribution of lamin B1 using super-resolution (STORM) imaging.

Discussion

Along with chromatin decompaction and relocation, the motion of genomic loci became more active in lamin B1-depleted cells. In control cells, mobility of the same chromosomal locus was found to be correlated with its subnuclear location.

SECTION 2: CONTENT FROM PAPER B

Source: uuid_0ba6a324-6c1f-1014-b700-b93a8e0baef4

H3.3 is essential for gastrulation of *Xenopus laevis*

Figure 1: H3.3 is essential for gastrulation of *Xenopus laevis*. The two well-characterized forms of non-centromeric H3 variants correspond to the replicative histones H3.1 and H3.2 and the non-replicative histone H3.3. In human, the replicative H3 form is comprised of H3.1 and H3.2 that differ by only one residue at position 96.

Figure 2: eH3.3 AIG single mutants rescue depletion of H3.3 during *Xenopus laevis* early development. Dedicated histone chaperones recognize histone variants by the H3.2 SVM and H3.3 AIG motifs. The morpholino against H3.3 induces gastrulation defects that are rescued by all eH3.3 with single mutations of the AIG motif.

Figure 3: eH3.3 AIG triple mutant rescues depletion of H3.3 during *Xenopus laevis* early development. In contrast, a triple mutant of H3.3 for the AIG motif, eH3.3 SVM, rescues the phenotype while carrying the same SVM motif than H3.2.

Figure 4: Swapping the AIG motif to the SVM motif leads to changes in chaperone interactions and histone modes of incorporation into chromatin in vivo. HIRA only recognizes the AIG motif while p60 recognizes solely the SVM motif.

Figure 5: H3.3S31 is critical *Xenopus laevis* early development and is phosphorylated in this model organism. Compared with eH3.3 WT, eH3.3 S31A mutant form cannot rescue the phenotype.

Figure 6: H3.3S31 negative charge is essential to rescue depletion of H3.3 during *Xenopus laevis* early development. In contrast to eH3.3 S31A, the use of an exogenous phosphomimic mutant, eH3.3 S31D, rescues the defects during gastrulation.

Figure 7: Graphical abstract. Defects associated with H3.3 depletion can be rescued with H3 histone variants carrying a potential negatively charge residue regardless of the mode of incorporation.

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FIGURES FROM PAPER B

Source: uuid_0ba6a324-6c1f-1014-b700-b93a8e0baef4

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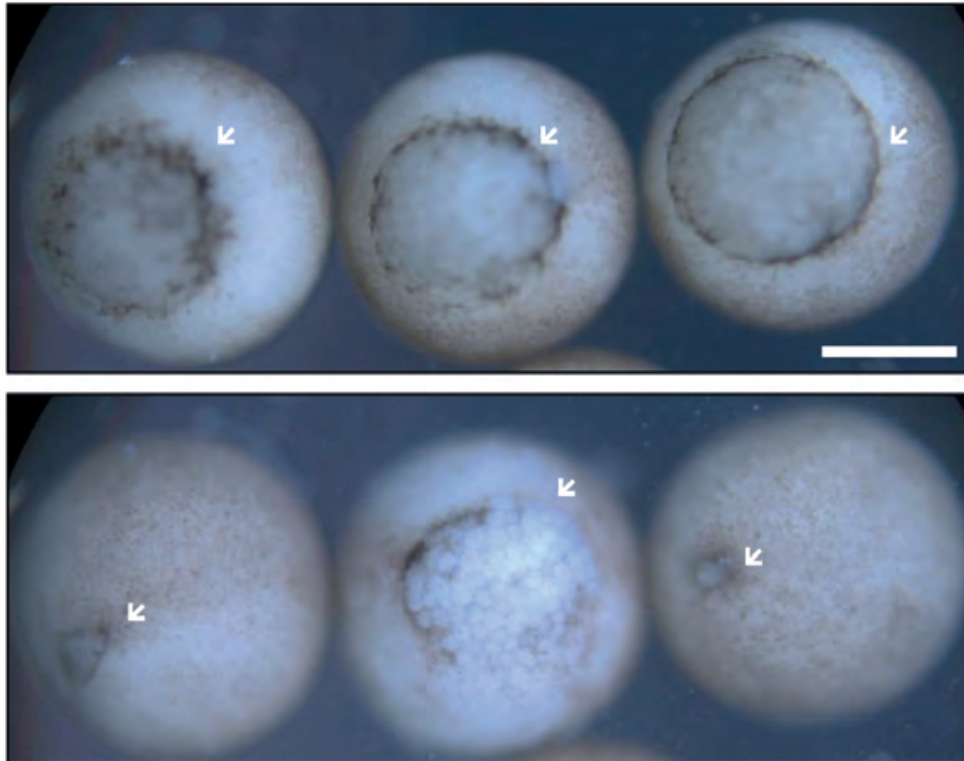


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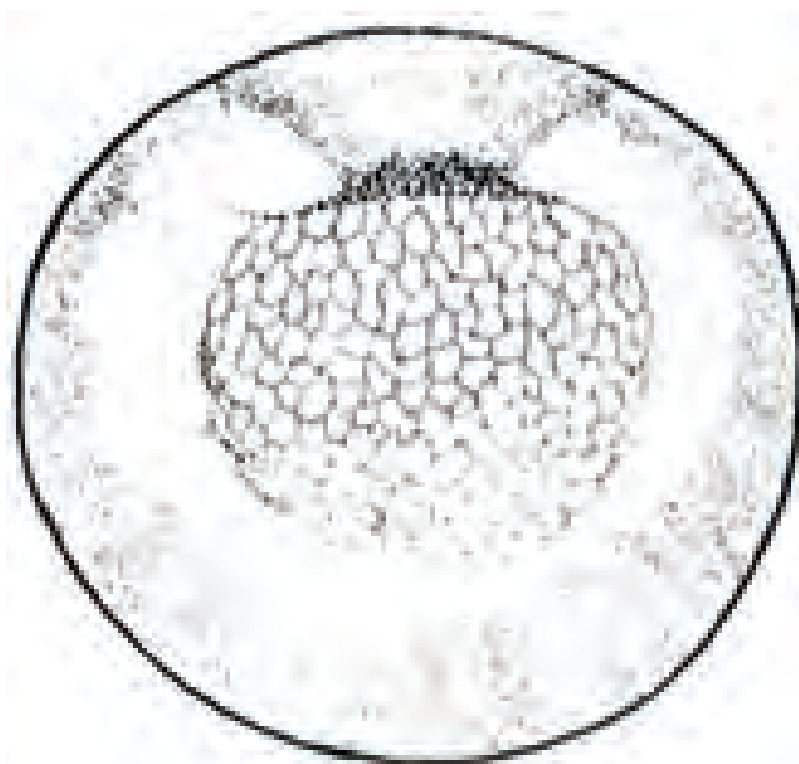


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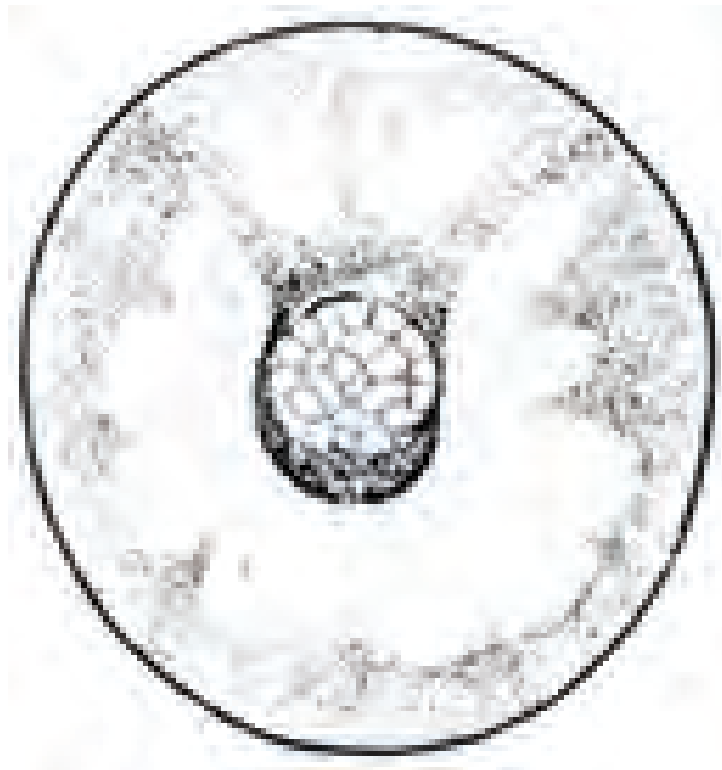


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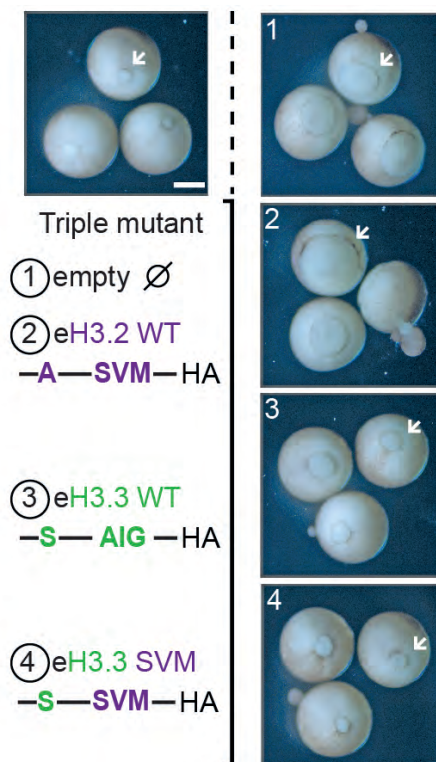


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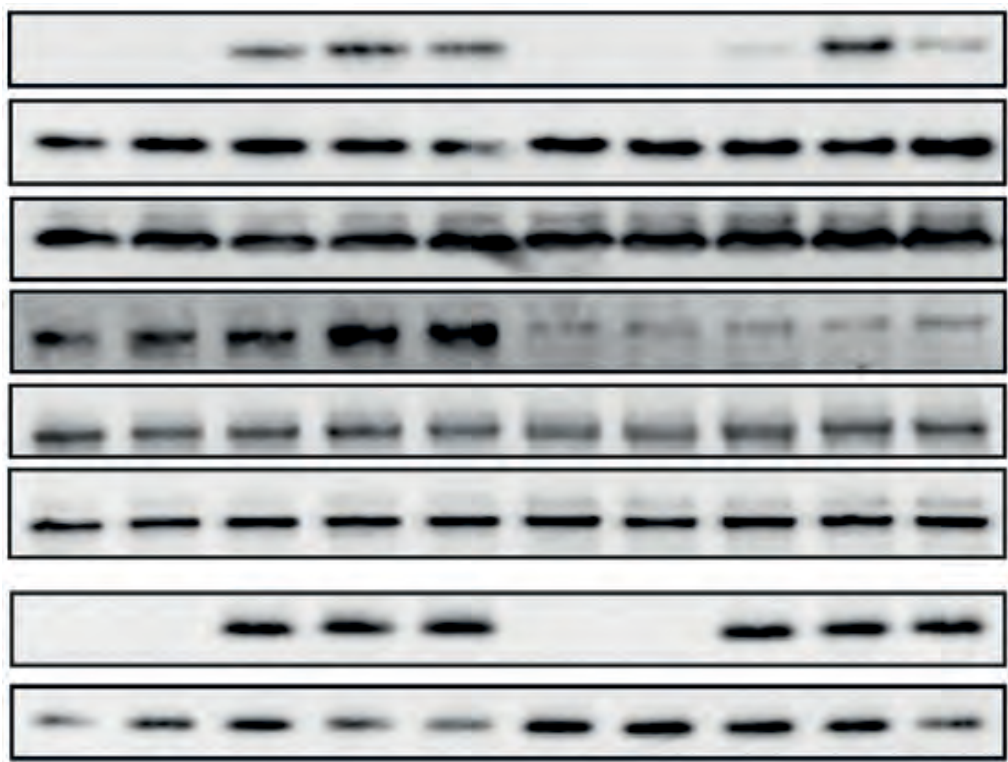


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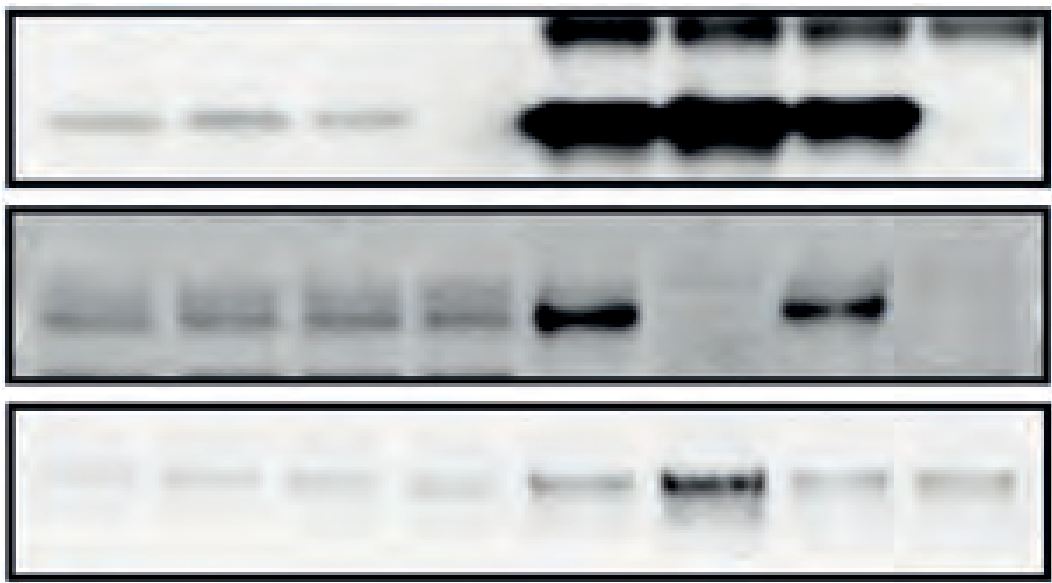


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Figure 7: fig_15.png

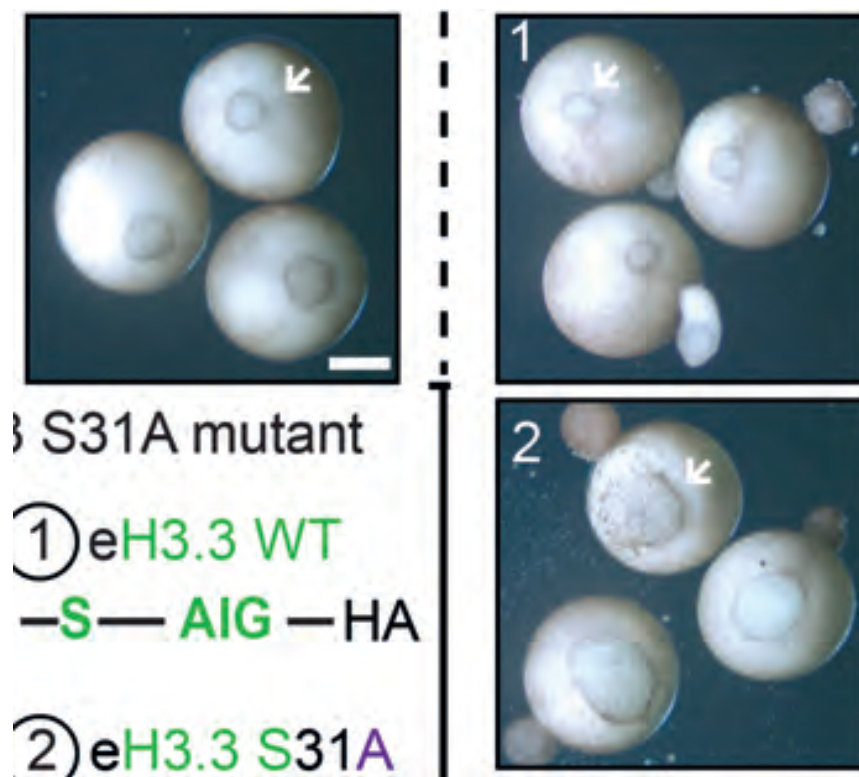


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Figure 9: fig_17.png

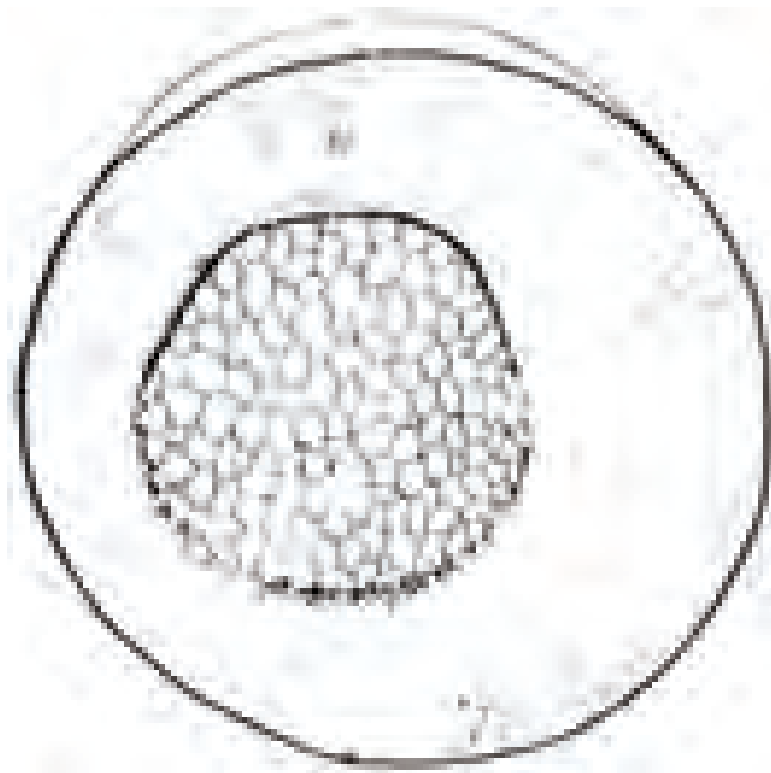


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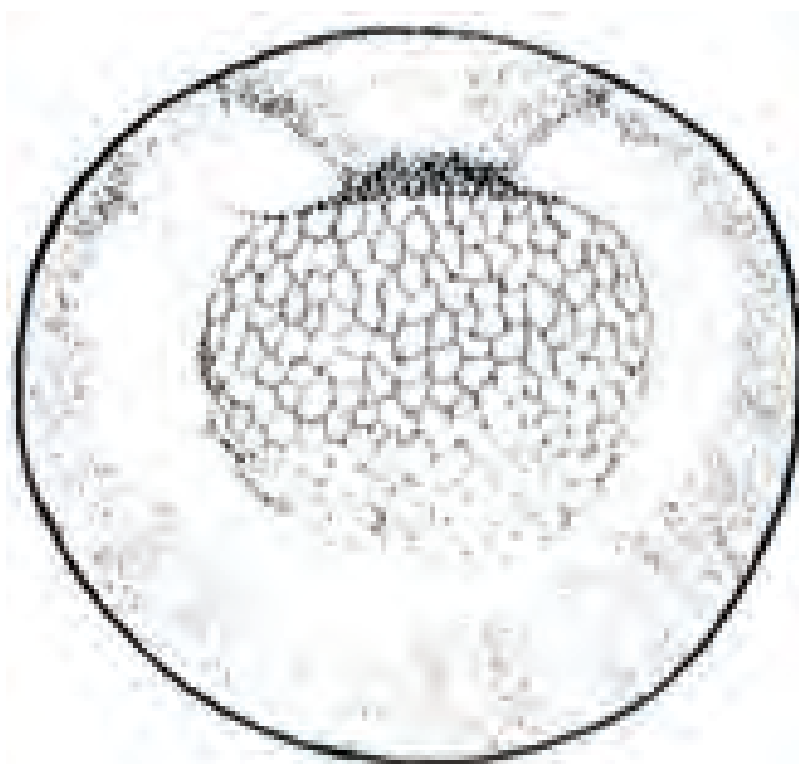


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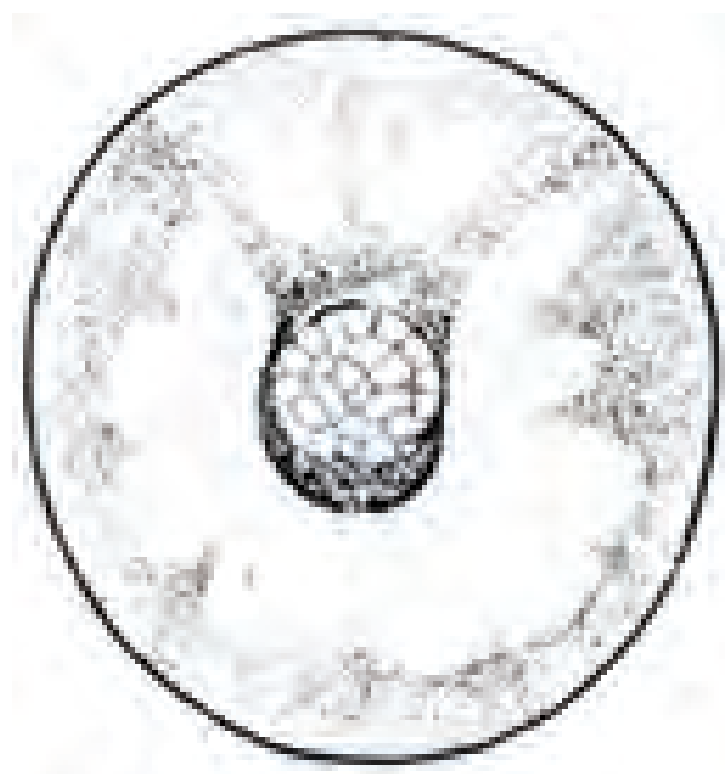


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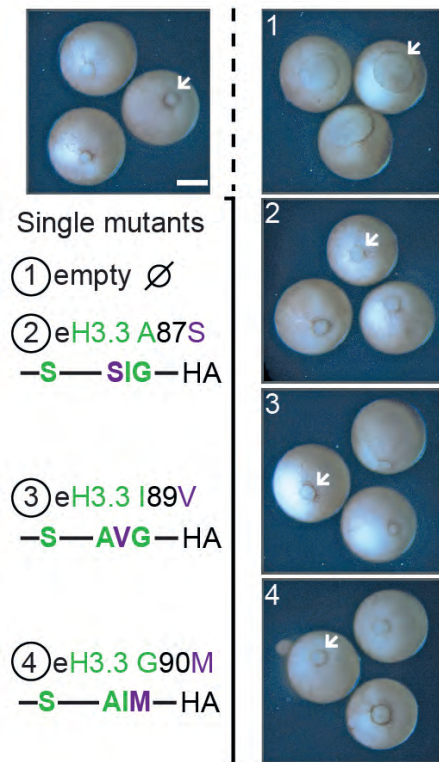


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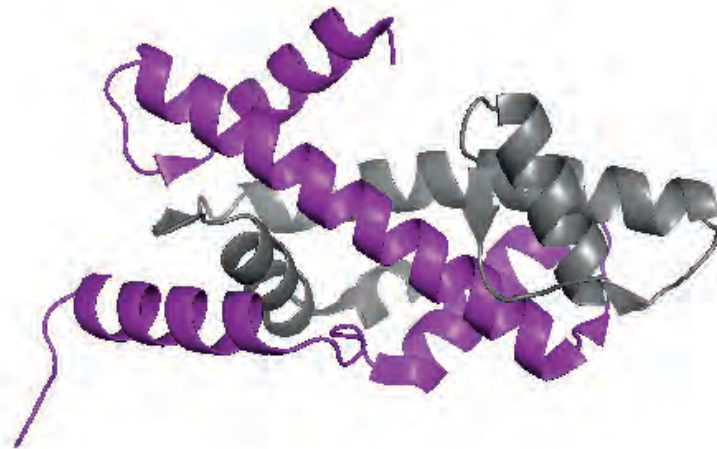


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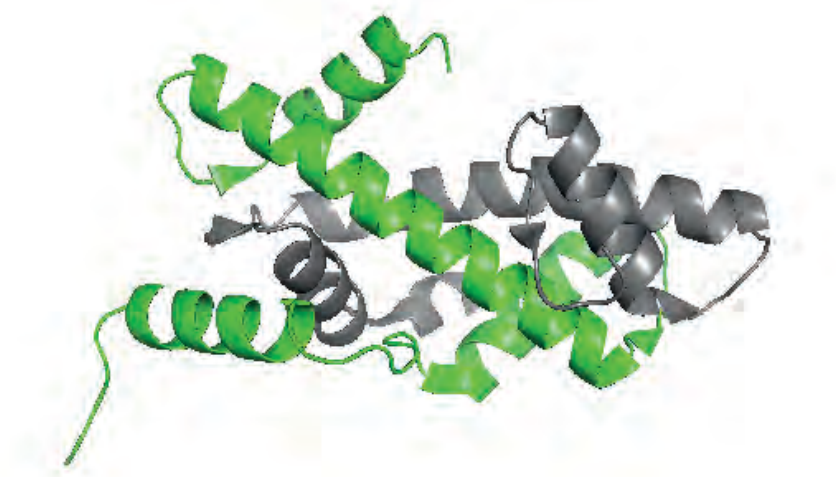


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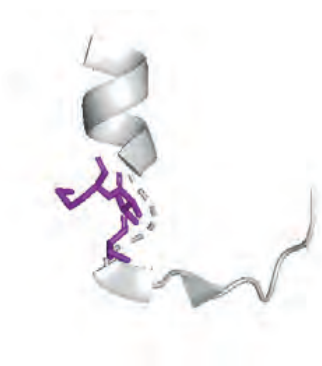


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Figure 17: fig_9.png